



Project : Expectation Decoder

→ 1. Understanding the Basics-

Q-1 What is probability?

→ Probability Means certainty of an event occurring

→ Probability values lies between 0 & 1

→ 0 → impossible

1 → certain

0.5 → 50% chance

⇒ From our dataset

Total student = 200

Student Passed ⇒ 70

$$P(\text{pass}) \Rightarrow \frac{70}{200} \Rightarrow 0.35 \Rightarrow 35\%$$

→ there are 35% chance that a student is passed out of 200 student.

Q-2 Key probability Terminology

→ Random experiment → An experiment is an action that produces an outcome.

→ in our dataset → Randomly selecting one student



2 outcome $\rightarrow$ The result of one trial
for ex $\rightarrow$ selected student $\rightarrow$ Pass or Fail.

3 Sample space $\rightarrow$ all possible outcomes.
 $\rightarrow$ checking if student is Pass or Fail

$$S = \{\text{Pass}, \text{Fail}\}$$

4 Event $\hat{=}$ An event is a specific outcome or group of outcomes we're interested in.

$\rightarrow$ ex: student passes the final exam.
we call this event A

$$P(A) \Rightarrow \frac{70}{200} \Rightarrow 0.35 \Rightarrow 35\%$$

$\rightarrow$ probability of event A

Q $\rightarrow$ Probability of Random variables

$\rightarrow$ Probability distribution of Random Variables

$\rightarrow$ as we have already interpreted RV in our ipynb file

$$\rightarrow RV = \{0, 1, 2, 3\}$$

$\rightarrow$ Probability for 0

$\rightarrow$ 0 $\rightarrow$ 'FFF'



$$\rightarrow X=0$$

one student Fail $\Rightarrow \frac{130}{200}$

$$\text{Second student Fail} = \frac{129}{199}$$

$$\text{Third student fail} = \frac{128}{198}$$

$$\rightarrow P(X=0) = \frac{130}{200} \times \frac{129}{199} \times \frac{128}{198}$$

$$P(X=0) \approx 0.65 \times 0.64 \times 0.646$$

$$P(X=0) \approx 0.27$$

$$\Rightarrow X=1 \rightarrow \text{one pass and Two fail.}$$

3 possibilities

PFF, FPF, FFP.

$$\rightarrow \text{PFF} \Rightarrow \frac{70}{200} \times \frac{130}{199} \times \frac{129}{198} \Rightarrow 0.148$$

Case 1

$$\rightarrow \text{FPF} \Rightarrow \frac{130}{200} \times \frac{70}{199} \times \frac{129}{198} \Rightarrow 0.149$$

Case 2

$$\rightarrow \text{FFP} \Rightarrow \frac{130}{200} \times \frac{129}{198} \times \frac{70}{198} \Rightarrow 0.149$$

Case 3

$$P(X=1) \Rightarrow \text{Case 1} + \text{Case 2} + \text{Case 3}$$



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$$P(X=1) \Rightarrow 0.446$$

$$\rightarrow P(X=2) \Rightarrow 0.239$$

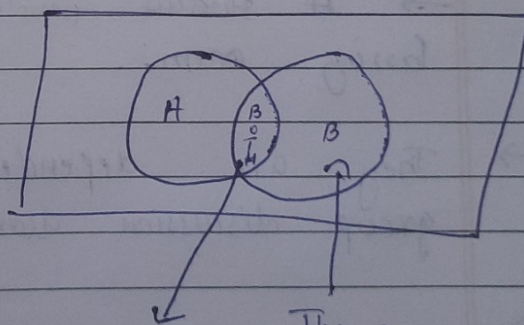
$$\rightarrow P(X=3) = 0.0417$$

$\rightarrow$ Probability distribution table

X	$P(X=x)$
0	0.27
1	0.446
2	0.239
3	0.0417

$\Rightarrow$ $\textcircled{Q} \rightarrow$ Intuition behind the conditional probability formula.

$$\rightarrow P(A|B) \Rightarrow \frac{P(A \cap B)}{P(B)}$$



$\rightarrow$ Think of event B as a filter. we stop looking at the whole pie and restrict universe only to 'B'. We then ask "out of these specific B situation what fraction also satisfies A ."



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→ identify if participating in group discussion and passing exam are independent, dependent or mutually exclusive event

⇒ event A → participating in group discussion
event B → passing exam

⇒ for independent situation $\rightarrow P(A|B) \Rightarrow P(A)$

$$\Rightarrow P(A|B) \Rightarrow \frac{40}{70} \Rightarrow 0.571 \quad \frac{40}{108} \Rightarrow 0.370$$

$$\Rightarrow P(A) \Rightarrow \frac{70}{200} \Rightarrow 0.35 \Rightarrow 0.34$$

→ so they are not independent event.

→ They are not mutually exclusive event cause both events can occur at same time.

→ A student involved in group discussion is passing exam.

→ They are dependent cause participating in group discussion change the probability of passing

→